

TEACHER GUIDE

Hose Reel Holder

Teaching guidance and complete student answer support

Use the companion as formative evidence

This guide accompanies the 50-page student workbook for the current Year 8 Stage 4 Hose Reel Holder course. The five modules span 20 weeks and 50 sixty-minute periods. The workbook supports this introductory unit within the wider Industrial Technology – Metal elective; it does not establish an assessment scheme for the whole course.

Use readings before the associated tasks. Model an observation or reasoning step, then ask students to apply it in their own words. The multiple-choice answers identify one best response. Open responses give expected evidence and acceptable alternatives rather than a single mandatory phrase. These are formative tasks; no marks or weighting are assigned here.

Practical information to provide

Supply the actual current plan, specifications and local workshop instructions. The course repository contains no authentic working drawing. Exact dimensions, stock, tolerances, machine settings, joint choices, coating products, curing conditions and test methods remain teacher-controlled. The book provides spaces to record this information; illustrations cannot supply it.

Teacher demonstrations, current school procedures, supervision and specific permission govern practical participation. A workbook response, safety poster or signed evidence entry does not itself authorise a machine process. Use the current assessment notification for dates, criteria, conditions and submission requirements.

Print and editable editions

Print the student PDF at A4 actual size, duplex on the long edge: 50 pages use 25 sheets. Word retains native headings, paragraphs, tables, questions and response rules. Editing may change pagination; check page references after local edits. Figures remain images, with native captions. No physical test print is claimed.

Support and extension

Allow students to rehearse an explanation orally, highlight the observation in a case, or label a sketch before writing. Extend by asking for an alternative explanation or a limit of the evidence. Keep the same technical and permission boundary for every learner.

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Map the learning and evidence

Use the page groups to locate the current course sections.

Course section	Student pages
M1 Function, users and constraints	4-6,9-10,12
M1 Workshop entry and machine permission	7-9,11-12
M1 Learn from the unsafe scenario	8-9,11
M2 Drawing to marked metal	13,15-17,19-20,46
M2 Use the design process	14,18,45
M2 Marking-out evidence	15-17,19-20,46
M3 Workpiece control	21,24-25,28
M3 Cutting and edge preparation	22,24-26
M3 Planned forming sequence	23,26-28,47
M4 Fit, alignment and finish	29-30,32-34,36
M4 Complete the approved joint	29,31-32,35-36
M4 Corrosion protection	30,32-36
M5 Evidence proves the process	37,39-41,44,47-48
M5 Organise the folio	38,42,44,50
M5 Evaluate with evidence	39-44,48-50

The twelve folio strands are distributed across the book: brief/users 4; risk 7-12, 25-28; research 10; ideas 18, 45; drawing 13-20, 46; materials/tools 6, 13, 21, 46; production planning 19, 26, 28, 46; mark/cut/form 20-28, 47; join/finish 29-36; testing/problems 39-44, 48; evaluation 49-50. Use attachments for additional authentic work rather than replacing the printed tasks.

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Answers for pages 3–6

Model reasoning and acceptable evidence

Student page 3 Use the book in the workshop

START-Q1 Record where you will keep your teacher-issued plan and assessment notification.

Accept a practical retrievable location or attachment arrangement. Check that the student distinguishes approved documents from illustrative workbook imagery.

Student page 4 Define the project brief

BRIEF-Q1 Who is the intended user, and what storage or access problem should the holder solve?

Look for an identified user, a specific storage/access need and a link to the outdoor context. Accept the class brief's actual user rather than requiring a fictional person.

BRIEF-Q2 Write two requirements. Explain why each matters to function or safe use.

Examples include stable support under approved normal use and convenient access to the hose. Require reasons; do not approve a load, mounting method or size from this response.

BRIEF-Q3 Record two constraints from the issued brief and one preference you may discuss with the teacher.

Constraints may include approved stock, tools, workshop time, issued dimensions or processes. A colour or appearance preference is acceptable only within teacher-approved choices.

BRIEF-Q4 Plan title or version and the detail you need clarified before making.

Student should identify the actual issued document and one genuine question, or record that the required details have been confirmed.

Student page 5 Design for a real user

M1-P5-Q1 Choose one visible feature. Explain how it could help a user, without assuming a dimension or test result.

Accept a visible supporting frame, clear access or other genuine feature connected to an intended benefit. The claim should be conditional; the image does not prove stability or load capacity.

Student page 6 Understand the steel you use

M1-P6-Q1 Explain why “mild steel” is not enough information to order the exact project material.

It identifies a material category but not the required stock form, dimensions, quantity or exact specification. These must come from the issued plan or teacher.

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Answers for pages 7–9

Model reasoning and acceptable evidence

Student page 7 Build a complete set of controls

M1-P7-Q1 Explain why wearing safety glasses does not make an unsecured drilling setup acceptable.

Glasses protect the eyes but do not prevent unsecured stock from spinning or moving. The student must stop and seek teacher direction so the approved workholding, guarding, training and permission are in place.

Student page 8 Read the whole unsafe scenario

M1-P8-Q1 Try a new case: a student has PPE and permission, but the work begins to vibrate differently. Explain the warning sign and the next response.

A change in vibration may indicate that control or the setup has changed. Stop as demonstrated, make the situation safe under teacher direction and report it. Earlier permission and PPE do not justify continuing after an unexpected change.

Student page 9 Check your starting decisions

M1-Q1 Which item should control the holder's exact dimensions?

B. The issued drawing carries the approved dimensions; appearance or convenience cannot replace it.

M1-Q2 Which statement is a design requirement rather than a preference?

C. Supporting the reel relates to essential function. The other statements express preferences.

M1-Q3 A guard is missing from the demonstrated setup. What is the best response?

D. A missing guard is a changed control. PPE and peer observation do not authorise a substitute setup.

M1-Q4 Which statement about stock is best supported?

C. The plan and teacher identify the required form and specification. Appearance or an unsupported general rule cannot do so.

M1-P9-Q5 Explain one answer in your own words, naming the reason rather than repeating the option.

Accept any correctly justified choice. Look for reference to authority, function, connected controls or material specification.

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Answers for pages 10–12

Model reasoning and acceptable evidence

Student page 10 Investigate a storage solution

M1-P10-Q1 Record the source of each example, then compare one useful feature. Explain which user need it addresses.

Expect identifiable sources, a visible feature from each example, and a reason linked to storage or access. A photograph cannot verify dimensions, durability, load capacity or an approved mounting method. This table and response are one research task.

M1-P10-Q2 Write one question you would ask before recommending either solution.

Accept a meaningful uncertainty such as actual user reach, suitability for the intended location, material approved by the teacher or evidence from an approved function check. Do not require a load test invented by the student.

Student page 11 Communicate one safety message

M1-P11-Q1 Design a safety poster for workshop entry. Include a clear message, a useful visual and the reason for the action.

Look for an accurate entry control such as clear walkways, required PPE or seeking permission; an unambiguous action; a relevant reason; legible original visual. Reject unsupported clearance distances, machine settings or claims that PPE alone is enough.

M1-P11-Q2 Explain how your words and visual help another Year 8 student act correctly.

Accept an explanation of audience, visibility, plain wording and how the visual reinforces the action. The reason must connect to the selected hazard, not simply say the poster looks good.

Student page 12 Record workshop readiness

M1-P12-Q1 Identify one user need and two constraints for your Hose Reel Holder. Explain how each could affect the finished project.

Example: the user needs stable, convenient hose storage. Outdoor exposure means the approved finish needs attention; available tools and time constrain the production choices. Accept other source-supported needs/constraints with clear cause and effect.

M1-P12-Q2 What does this readiness record show, and what still requires the teacher's decision?

It records the student's understanding and observed preparation. It does not itself grant machine permission or certify competence. Teacher direction and current school procedures still govern practical participation.

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Answers for pages 13–16

Model reasoning and acceptable evidence

Student page 13 Read the plan before the metal

M2-P13-Q1 Name one detail you must obtain from the issued plan rather than estimate from a photograph.

Accept a written dimension, material specification, exact component shape, feature location, tolerance or production note. A useful answer identifies a consequential detail that appearance alone cannot verify.

Student page 14 Develop an idea with reasons

M2-P14-Q1 Rewrite “I chose it because it looks best” as a reason linked to a user need.

Example: I chose this arrangement because its open access would make the hose easier for the intended user to place and remove. Accept another specific criterion and a plausible link to the idea; do not require an untested claim to be presented as a proven result.

Student page 15 One reliable starting point

M2-P15-Q1 Explain why a common datum and a separate check work together to improve accuracy.

A common datum makes each location depend on one consistent reference rather than on preceding marks. A separate check against the issued drawing can catch a wrong reference, misread position or unclear line. Neither step guarantees accuracy by itself; both support a reliable pre-cut decision.

Student page 16 Catch the error while it is reversible

M2-P16-Q1 Your marked part looks correct, but you cannot identify which edge was used as the datum. Explain your next steps and the evidence you would record before cutting.

Pause the cut; locate the issued drawing and identify the approved datum and relevant face. Compare the marked part with those references, seek teacher clarification if the starting edge cannot be established, and use the approved process to resolve any ambiguity. Recheck before proceeding. Record a labelled view or sketch that identifies the confirmed datum, the checked marks and the decision; do not invent approval or retrospectively claim an unrecorded check.

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Answers for pages 17–18

Model reasoning and acceptable evidence

Student page 17 Check your planning decisions

M2-P17-Q1 A printed drawing has been reduced to fit a page. Which information should control the size of the part?

B — Written dimensions on the approved drawing control size. Resizing a page changes the scale of its outline; a photograph or convenient stock length is not construction authority.

M2-P17-Q2 Which record gives the strongest reason for choosing an idea?

D — The reason connects a design choice with a user need that can be checked. The other options state preference, imitation or an unsupported impression.

M2-P17-Q3 A student measures each new position from the mark just made. What is the main concern?

A — A chain of measurements can carry an earlier error into later positions. A common approved datum reduces that dependency; the other statements do not describe the effect.

M2-P17-Q4 Which photograph would best support a pre-cut marking check?

C — The photograph connects the controlling information with the reference and the feature checked. Its caption should state what was checked and the result; a photograph alone cannot prove every check happened.

Student page 18 Two ideas, one need

M2-P18-D1 Concept A: sketch an arrangement and annotate the feature that is intended to help the user.

Look for a legible concept showing where the hose is stored and how the user could access it. At least one annotation must connect a feature to access or organisation. No unsupported project dimensions or mounting details are required.

M2-P18-D2 Concept B: change one important design decision and annotate its intended benefit.

The second concept should change a meaningful relationship, such as access direction, arrangement or space use, rather than colour alone. Accept multiple ideas. An annotation should explain the intended benefit without claiming untested performance.

M2-P18-Q1 Compare both ideas using one shared criterion. Which would you explore further, and why?

The response names one shared criterion, compares both concepts against it and explains a provisional choice. Accept either concept when the reason is consistent with the sketches; useful answers acknowledge a remaining question for investigation or teacher review.

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Answers for pages 19–20

Model reasoning and acceptable evidence

Student page 19 Rebuild a reliable mark-out check

M2-P19-Q1 Write your order. Explain why two of the actions must occur before a later action.

A defensible order is D, B, F, E, A, C. F can occur immediately after D if the uncertainty must be resolved before the datum or face can be identified. Reading and resolving uncertainty precede marking; identifying the reference precedes locating marks; checking follows marking; the record follows the check it reports. Accept another order only if it preserves these dependencies and explains them.

M2-P19-Q2 Which action would you repeat if the teacher issued a revised drawing before cutting? Explain.

Repeat the drawing review, then confirm the affected part, face, datum and marks against the revision. Resolve changes, recheck the mark-out and update the evidence record before cutting. A revised drawing cannot be treated as equivalent without checking what changed.

Student page 20 Make the marking check visible

M2-P20-D1 Record the marked part. Label the datum, one checked feature and the intended waste area where relevant.

Expected evidence is the student's own clear sketch or permitted photograph, with an identifiable datum and feature. Waste is marked where relevant to the checked operation. The record must be consistent with the issued drawing; a planning-only sketch is acceptable when accurately labelled and must not claim a completed practical check.

M2-P20-Q1 Plan or revision used, check completed, and result:

The student identifies the teacher-issued plan or revision sufficiently to distinguish it, states the feature and reference checked, and reports the actual result. A result may include an error or request for clarification; correctness should not be fabricated.

M2-P20-Q2 Explain how the reference edge and a second measurement check improved, or will improve, the accuracy of your marked metal.

This is an original adaptation of the module written-response purpose. The explanation links a consistent starting edge to comparable feature locations, and a second check against the issued plan to detecting wrong positions, unclear marks or the wrong reference. Use future tense for planning-only work and actual evidence for completed work.

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Answers for pages 21–24

Model reasoning and acceptable evidence

Student page 21 Control begins with the workpiece

M3-P21-Q1 Explain one way secure workholding can improve accuracy as well as reduce risk.

Example: limiting unwanted movement helps keep the marked feature in its intended position relative to the tool and reduces the chance that the part will slip or spin. Accept another clear connection between restraint/support, controlled position and a safer or more accurate result.

Student page 22 A cut has an edge as well as a shape

M3-P22-Q1 Explain why “the part is cut to shape” is not enough evidence that it is ready for assembly.

A cut can leave burrs, sharp edges or uneven areas. These can affect safe handling, fit and surface preparation. The student should identify the need for the teacher-approved edge inspection and treatment, then a check that the required geometry remains correct. Do not accept finger testing, bare-hand swarf clearing or arbitrary removal of extra metal.

Student page 23 Forming changes the next decision

M3-P23-Q1 Name one way forming could affect a later operation.

Accept a plausible effect such as changing tool access, the direction of a hole or face, component alignment or the ability to hold the part. The response should connect the changed shape to a later decision without prescribing an unsupported production sequence.

Student page 24 Notice the change, explain the response

M3-P24-Q1 A forming trial leaves a feature pointing in an unexpected direction. Explain what you would report, what you would avoid doing, and what must be confirmed before continuing.

Report the observed feature and unexpected direction, identifying the relevant plan or reference and stage. Stop the affected work; avoid applying extra force, changing the sequence or improvising a correction. The teacher must help establish the cause and confirm the approved correction, setup and next step. Record the actual check and decision. Accept clear alternatives that preserve observation, teacher control and an honest evidence trail.

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Answers for pages 25–26

Model reasoning and acceptable evidence

Student page 25 Choose a controlled response

M3-P25-Q1 A marked part can still rotate in its proposed holding arrangement. What should happen next?

C — The actual part must be secured through the approved setup before work starts. Slower work, hand holding and extra marks do not resolve uncontrolled rotation.

M3-P25-Q2 Which statement best describes a burr?

A — A burr is an unwanted edge projection. It can affect safe handling, fit and finish; it is not a drawing feature, coating or datum.

M3-P25-Q3 A new sound occurs during a practical operation. Which response best follows the learning?

B — A changed sound can signal a developing problem. The student should use the demonstrated safe stop and report the observation rather than continue or improvise a correction.

M3-P25-Q4 Why does a forming sequence need planning?

D — Forming can change the orientation, alignment and accessibility of features. That relationship makes the approved sequence and trial checks important.

Student page 26 Read the warning signs

M3-P26-Q1 Case A — concern and response:

Increased vibration is a changed condition and may indicate loss of control. Stop safely using the demonstrated method, keep clear and tell the teacher what changed. The cause and approved correction must be checked; do not push harder or hold the moving work.

M3-P26-Q2 Case B — concern and response:

The projection may be a sharp burr; finger testing can cause injury. Use the approved visual/handling inspection and teacher-demonstrated edge-treatment process. Keep the work controlled and recheck against the required geometry.

M3-P26-Q3 Case C — concern and response:

Misalignment shows a difference from the reference, but its cause is not yet known. Pause, identify the face, direction and stage, and ask the teacher to check the plan and approved correction. Further uncontrolled force could worsen the error or introduce risk.

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Answers for page 27

Model reasoning and acceptable evidence

Student page 27 Show what your check needs to reveal

M3-P27-D1 Sketch a cut edge before and after approved preparation. Use labels to distinguish the required edge from an unwanted projection.

The paired conceptual drawings should show an unwanted burr or projection and a prepared edge that preserves the intended basic geometry. Labels distinguish the defect from the required part. No finger inspection or excessive reshaping should be implied.

M3-P27-D2 Sketch the evidence view you would use to show a formed feature against its approved reference. Label the feature, reference and comparison.

Look for a view in which both the formed feature and an approved reference could be compared. Labels identify what is being checked and where the reference comes from. The student is planning evidence, not fabricating a project drawing; exact geometry and dimensions remain teacher-controlled.

M3-P27-Q1 What detail would be lost if your evidence picture were taken too close?

Accept a specific loss such as the reference, orientation, overall alignment or the connection between the checked feature and the rest of the part. The answer should link framing to whether the intended comparison can be understood.

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Answers for pages 28–30

Model reasoning and acceptable evidence

Student page 28 Record a controlled production decision

M3-P28-D1 Sketch or attach permitted evidence of the checked feature. Label what the image proves.

The evidence should be the student's own, identify a real stage and show the relevant checked feature or teacher-approved reference. A planning-only record is acceptable if clearly labelled. The picture should not be used as proof of a hidden check or an unobserved machine setup.

M3-P28-Q1 Explain how secure work, a balanced body position and stopping when conditions change reduce risk. Link at least one point to your record.

This preserves the purpose of the module written response. Secure support/restraint limits slipping, spinning, vibration or falling. The demonstrated balanced position supports control and keeps the body clear of the likely tool path. A changed condition requires a safe stop and teacher report before continuing. The student links one principle to actual evidence or explicitly describes it as a plan.

M3-P28-T1 Complete the production record above using actual evidence or clearly labelled planning.

Entries identify the stage and sequence, the approved source/demonstration, and a specific check with an honest result. Teacher confirmation is required for exact practical details. Accept an unresolved result if the next action is recorded; do not reward invented approval.

Student page 29 Fit first, then fix

M4-P29-Q1 Why can a correctly shaped individual part still create an assembly problem?

It may be in the wrong orientation or not align/fit with another component as intended; an individual outline does not prove the relationship between assembled parts. Accept another relevant relational explanation, with the issued plan as the checking reference.

Student page 30 Keep the protection continuous

M4-P30-Q1 A student says, "The colour is even, so the holder will last outdoors." Explain what this statement overlooks.

Even colour alone does not establish preparation quality, complete coverage, correct product choice/application/curing or long-term durability. The student should connect outdoor corrosion protection with a safe clean prepared surface and teacher-approved product instructions. Accept discussion of damage or exposed areas; do not require an invented service life or test result.

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Answers for pages 31–33

Model reasoning and acceptable evidence

Student page 31 The joint belongs to the design

M4-P31-Q1 What is the difference between knowing a joining method exists and being authorised to use it?

Knowing a method exists is conceptual knowledge. Authorisation requires the teacher-issued plan/demonstration and relevant permission, setup, supervision and controls for the actual classroom task. The student cannot select or perform a method merely because it is familiar.

Student page 32 A mismatch is information

M4-P32-Q1 A trial assembly lines up, but a contact surface still has visible contamination. Explain why the project is not ready to proceed, and describe the evidence needed after the issue is resolved.

Correct alignment addresses only one requirement. Contamination can affect the approved joining/finishing preparation and must be handled through the demonstrated process. Pause the affected step, report the surface condition, obtain the teacher-controlled preparation/correction and recheck. Record the relevant before/after observation and actual result; do not claim a particular cleaning chemical, joining method or durability result without authority.

Student page 33 Build a quality pathway

M4-P33-Q1 For A–D, state what needs checking or confirming first.

A: compare fit/orientation/alignment with the issued plan and have the teacher confirm correction before joining. B: complete the relevant post-joining assembly check before claiming correctness. C: complete the teacher-approved surface preparation before coating. D: obtain and follow actual product instructions and teacher direction for curing and readiness; do not invent a time.

M4-P33-Q2 Choose one situation. Explain how an early check protects the quality of a later stage.

Accept a causal explanation tied to one case. Example: resolving a trial-fit mismatch before joining avoids locking parts into the wrong relationship and allows a clearer correction. Another example links preparation to effective coating coverage or product curing requirements to readiness for handling/testing. A slogan such as “checks make it better” is insufficient without the relationship.

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Answers for pages 34–35

Model reasoning and acceptable evidence

Student page 34 Check fit, joining and protection

M4-P34-Q1 Which observation is most useful during a dry check?

C — A dry check examines the intended fit, orientation and alignment before permanent joining. The other observations do not establish the relationship between the components.

M4-P34-Q2 A student finds a quicker joining method online. What should determine the method used in class?

A — The controlling plan and teacher-approved process determine the joint. Online popularity or convenience cannot change the project authority or remove preparation requirements.

M4-P34-Q3 Why is careful surface preparation important before protection?

D — Safe, clean, properly prepared surfaces support the protective finish. Preparation alone does not choose a product, prove function or determine whether a joint is removable.

M4-P34-Q4 A coating looks dry, but its curing requirements are unknown. What is the best next step?

B — The actual product instructions and teacher direction control curing and readiness. Appearance or memory cannot establish the correct requirements.

Student page 35 Make the comparison visible

M4-P35-D1 Draw a trial fit with one possible alignment problem. Label the intended meeting feature, observed difference and reference needed.

Look for a plausible mismatch between meeting edges or corresponding features, labelled separately from the intended relationship. The student identifies the issued plan or teacher-approved reference needed to judge it. Do not require exact holder geometry or accept a forced correction as the answer.

M4-P35-D2 Draw a close detail of an outdoor steel surface with one area needing attention before protection. Annotate why the detail matters.

Accept a burr, dirt, scale, contamination or exposed area, clearly labelled and connected to safe preparation or complete protection. The student should not prescribe an unsupported chemical/process or claim that a decorative colour proves durability.

M4-P35-Q1 How do the two drawings show different kinds of quality?

The first checks the relationship of parts: fit, orientation or alignment. The second checks surface condition and readiness for protection. Both matter because an assembly can fit but have poor preparation, or have an attractive surface but incorrect alignment. Accept equivalent reasoning.

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Answers for pages 36–38

Model reasoning and acceptable evidence

Student page 36 Connect the checks to the finish

M4-P36-D1 Sketch or attach permitted evidence of one completed check. Label what the detail proves.

The evidence depicts the student's own assembly or surface check and labels the relevant feature, observation and source/reference. A planned view is acceptable only if labelled as planned. It cannot prove hidden joint strength or long-term protection from appearance alone.

M4-P36-Q1 Explain how trial fitting, careful alignment and surface preparation improve the quality and durability of the Hose Reel Holder.

This adapts the module written-response purpose. Trial fitting reveals fit/orientation problems before permanent joining; careful alignment helps components sit and function as intended; safe, clean, correctly prepared surfaces support the teacher-approved corrosion protection. Answers should link causes to outcomes and distinguish observed quality from predicted durability. Exact project/process details remain teacher-confirmed.

M4-P36-T1 Complete the quality record with actual evidence, sources or an honest “not yet completed”.

Check that each row names the relevant feature, observed result or source. The joint and finish must match the teacher-controlled information. A product name should come from the classroom label or supplied information, with no invented curing time or authorisation.

Student page 37 Evidence shows the thinking

M5-P37-Q1 Name one decision a final product photograph alone would not reliably prove.

Accept a pre-cut check, source/reference used, reason for choosing an idea, approved correction, safe stop or actual test result. The answer should identify information about process or reasoning that is not directly established by the final appearance.

Student page 38 Give the folio a clear route

M5-P38-Q1 Two photos show the same stage from almost the same angle. When might keeping both be useful?

Keep both only if each reveals a distinct relevant detail, comparison, change or result that one cannot show alone. The explanation should identify that difference; more photos without new evidence are not automatically better.

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Answers for pages 39–41

Model reasoning and acceptable evidence

Student page 39 Connect a claim to its evidence

M5-P39-Q1 Explain the difference between “the finish is even” and “the holder will never corrode”. What extra issue does the second claim raise?

The first is a limited observation about visible surface appearance. The second is an absolute long-term durability claim that the observation cannot establish. It would require relevant evidence over appropriate conditions/time and cannot be proved by appearance alone. A suitable evaluation reports visible condition and follows teacher-approved criteria without promising that corrosion can never occur.

Student page 40 Make the judgement precise

M5-P40-Q1 Write a more precise version of this claim: “My folio proves that every stage was accurate.” The record contains a clear marking check but no evidence of the later assembly check.

Example: My folio shows that I checked the marked positions against the issued plan before cutting. It does not yet provide evidence of the assembly check, so I cannot use it to claim that every stage was accurate. I need the actual assembly record or must state that limitation. Accept equivalent wording that limits the claim, identifies the missing evidence and avoids inventing a check.

Student page 41 Judge the strength of the evidence

M5-P41-Q1 Which caption provides the most useful process evidence?

D — It identifies a stage, reference and check with a recorded result. The teacher should still verify that the evidence supports the claim. The other captions provide opinion or irrelevant information.

M5-P41-Q2 Which record should control a functional test?

B — Testing follows teacher-approved criteria and instructions. The workbook does not supply independent loads, methods or permission to improvise a test.

M5-P41-Q3 A close-up shows an even surface finish. Which claim stays closest to that evidence?

A — The image supports a limited visible observation. Hidden joint performance, future corrosion and the entire process sequence require different evidence.

M5-P41-Q4 Which improvement is the most specific and useful?

C — It identifies a practical change to evidence timing and explains the intended benefit. The others are vague, quantity-focused or conceal useful process evidence.

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Answers for pages 42–43

Model reasoning and acceptable evidence

Student page 42 Match the evidence to the claim

M5-P42-Q1 Choose the best record for each claim: 1. I compared design options. 2. I checked the marks before cutting. 3. I responded to a changed condition. 4. I evaluated the result against an approved criterion.

1–A; 2–B; 3–C; 4–D. A links options to a shared need; B connects datum, marked features and a recorded check; C describes an observed change and controlled response; D connects a test observation to an approved criterion. E alone is weaker evidence of these process claims.

M5-P42-Q2 Which claim could record E support? Explain one important limit.

Example: E can support a limited claim about the visible finished form or appearance shown in the photograph. It cannot alone prove hidden joint quality, the production sequence, safe practice, testing results or long-term durability. Accept another directly visible claim with a clearly stated limit.

Student page 43 Evaluate without inventing a result

M5-P43-Q1 Write an evidence-based evaluation. Include one supported strength, one concern and one conclusion that cannot yet be made.

A sound response states that the trial-fit record supports alignment of the checked features, while the exposed patch is a concern for complete protection and needs teacher-approved attention. The absent functional test means the student cannot yet claim that function met the approved criteria or that the whole product is a complete success. Do not require a single wording; reward explicit links to the provided evidence and its limits.

M5-P43-Q2 Propose one realistic improvement to the student's process or evidence record. Explain its likely benefit and what must be confirmed before action.

Accept a planned systematic surface check, timely completion/documentation of the teacher-approved functional test, or a clearer evidence checklist that identifies missing stages. The student should explain how the change addresses a stated concern. Practical correction/testing must follow teacher-approved instructions and readiness requirements; no invented loads, product times or claimed completed result. A folio-only improvement may identify the teacher's required evidence/format as the confirmation.

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Answers for pages 44–45

Model reasoning and acceptable evidence

Student page 44 Make your final reflection traceable

M5-P44-D1 Place or sketch your chosen evidence. Add its stage and date or sequence label.

Expected evidence is the student's own identifiable project record, with a stage and date/sequence label. It should be sufficiently clear to support the later explanation. Planning-only evidence must be labelled honestly; generated examples or another student's work cannot be passed off as the student's own result.

M5-P44-Q1 What decision or check does the evidence show? Explain its result and one limit on what it proves.

The response identifies the actual decision/check, relevant reference or criterion and observed result. It states a sensible limit, such as not proving hidden strength, later accuracy or long-term durability. For planning-only work, expected observations may be described but cannot be reported as completed results.

M5-P44-Q2 Name one realistic improvement for future work and explain why it would help.

This adapts the module written-response purpose. Accept a specific improvement to a design decision, production sequence, checking point, skill practice or evidence timing, justified by the chosen record. It should be feasible within teacher-approved classroom processes, with relevant confirmation noted. "Try harder" alone is insufficient; a good answer connects an action to a safer, more accurate, more durable or clearer result.

Student page 45 Explore two permitted options

F45-Q1 Option A — sketch the whole idea and annotate a feature, its intended benefit and a constraint.

Look for a complete understandable concept with a functional annotation and a real constraint. Any altered feature remains a proposal pending approval; no invented dimension should be treated as a construction instruction.

F45-Q2 Option B — show a meaningful alternative and explain how its effect differs.

A valid alternative changes an allowed feature or approach in a way that affects a criterion. A duplicate sketch with no meaningful explanation is weak evidence.

F45-Q3 Compare the options against one shared criterion and record the teacher feedback needed.

Both options should be compared against the same relevant criterion using reasons. Require teacher feedback before a proposal changes the issued construction plan.

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Answers for pages 46–48

Model reasoning and acceptable evidence

Student page 46 Keep the approved plan traceable

F46-Q1 Add a reference sketch or attach a permitted copy of the issued plan. Identify the datum and one feature to check. Use written dimensions from that plan.

Check that the student refers to an authentic issued plan and identifies a correct datum/feature within it. A copied or attached drawing should remain legible; this workbook field is not a source of dimensions. The document table forms part of the same task.

F46-Q2 Explain one check you will complete before the next irreversible step.

Expected evidence names a specific feature, the approved information it is checked against and when the check occurs. Accept accurate plan-specific alternatives; a teacher must resolve unknown practical details.

Student page 47 Keep a useful production record

F47-Q1 Choose the entry that best shows your learning. Explain the evidence and what you changed or confirmed.

The table and response should form a chronological evidence trail. Look for a particular decision/check, an observed result and the approved next step. A problem can be strong evidence when its response is accurately recorded. No invented teacher approval or result.

Student page 48 Record an approved test

F48-Q1 Explain what this test establishes and one limit of the evidence.

Require a real teacher-approved method, truthful result and bounded conclusion. A short function check may show fit/access under those conditions; it cannot establish long-term corrosion resistance or an unspecified maximum load. If no test occurred, credit a clear pending record rather than an invented pass.

F48-Q2 Record the teacher-directed next step if the result differs from the criterion.

Accept a specific authorised next step or a question awaiting direction. Do not accept an unapproved repeat with greater load or a concealed failure.

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Answers for pages 49–50

Model reasoning and acceptable evidence

Student page 49 Evaluate the project with evidence

F49-Q1 Function — explain how well the holder meets the intended user need. Refer to one recorded check.

Look for a criterion, an actual observation and a justified conclusion. Where testing is incomplete, the judgement must state that limitation instead of claiming success.

F49-Q2 Quality — discuss one strength and one limitation in accuracy, alignment, edge preparation or finish.

Accept source-appropriate quality observations linked to real work. Require specific evidence rather than “good” or “bad”. A photo can support an appearance observation but may not prove hidden quality or durability.

F49-Q3 Improvement — propose one realistic change to a decision, checking point or skill practice. Explain its likely effect.

A sound improvement responds to the identified limitation, is feasible within teacher-approved boundaries and explains a predicted benefit. It must not be presented as a proven result or permission to alter the plan.

Student page 50 Present a clear evidence trail

F50-Q1 Check the index, then explain one skill or idea you can now use more confidently. Refer to evidence that shows the change.

Expect a completed, accurate evidence index and a reflection linked to a particular entry or comparison. Valid learning includes datum use, distinguishing observation from a claim, recognising changed controls or writing an evidence-based evaluation.

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Source and image notes

Source snapshot 12 September 2026

Current course

Hose Reel Holder guided course: <https://stevenowell.github.io/Yr-8-Metal-Technology/> . The current course contract and guided/data.js control identity, scope and all 15 named sections. Repository snapshot 32d44bf9dacb9a4e0bb1dd67afe50aba8d3726b6. The current twelve-card folio supplies the evidence sequence.

The 50-period Teaching Program v3.0 records the 2027 course pathway and existing syllabus decision. This workbook does not create new outcome mapping. NSW Industrial Technology 7–10 Syllabus 2025: <https://curriculum.nsw.edu.au/learning-areas/tas/industrial-technology-7-10-2025/overview/course> .

Bounded enrichment

The older teacher document Hose Holder Program Theory Package.docx (provider modified 20 March 2023) supports broad enrichment on tools, materials, safety communication and project records. Its readable connector text was inspected and retained. Original document bytes and embedded images were not available for local verification; they were not reproduced.

Conflicts resolved

Use the current 20-week course pacing; older slides retain stale two-week module labels. Historical assessment weights and older practical scenarios are excluded as authority. The book does not import numerical guard clearances, workpiece dimensions, machine settings, joining sequences or load tests from those activities.

Figures

The cover, safety still life, steel stock, corrosion, edge, trial-fit, forming and evidence photographs were created using the built-in image generation tool for this workbook. Its exact model version was not exposed. The datum diagram is an original conceptual schematic. The landscape holder image comes from the user's current course and remains illustrative. Exact prompts, source bytes and hashes are retained with the build.

Each reused image has a stated local teaching purpose in its caption. Generated and course illustrations are not authenticated student work, construction plans or safety certification. Figure captions remain editable in Word. The Mirror Edition 2.0, Footstool and Desk Tidy benchmarks guide editorial design; their unit facts are not used here.